BIOÉNERGÉTICS

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I - Introduction

II- Basal Metabolism (BM)

III- Thermogenesis

IV- Physical Activity

V- Balance of Energy Intake and Expenditure

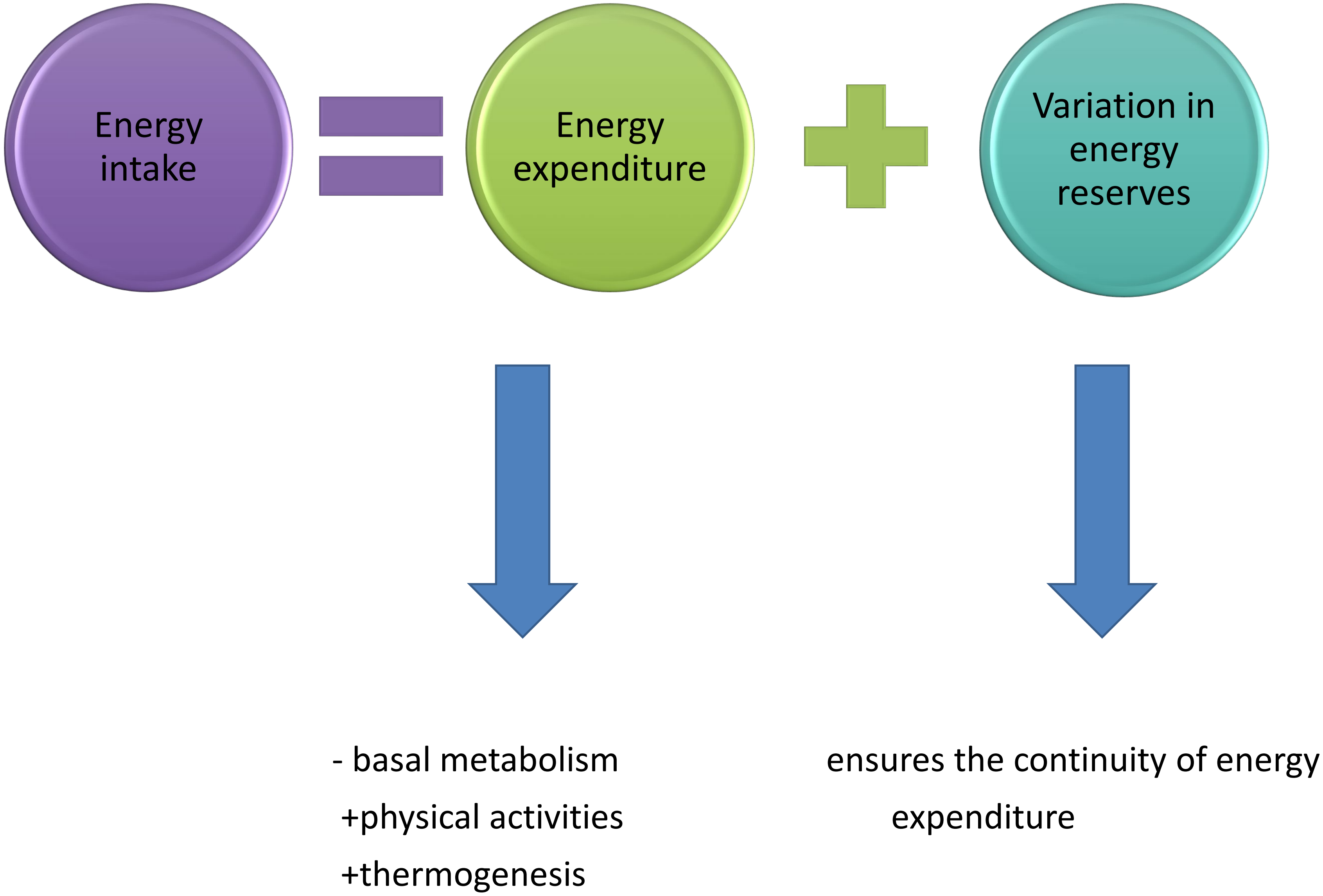
VI- Measurement Methods (Calorimetry)

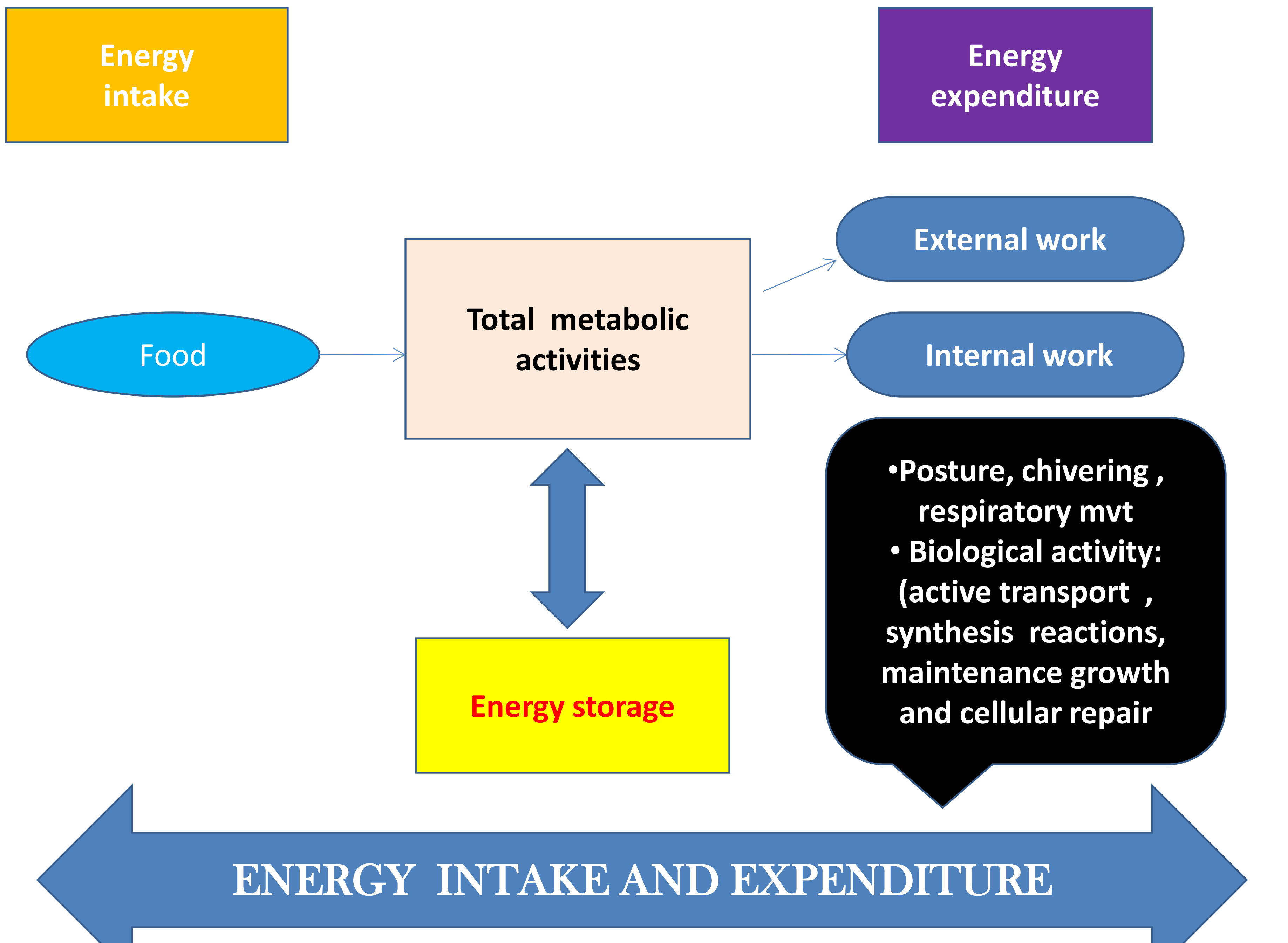
VII- Food Ration

I- Introduction
Définitions :

Bioenergetics : origin and fate of energy in living matter

Man : a heterotrophic being (uses chemical energy contained in carbohydrates, lipids, and proteins (C.L.P)) Feeding (plants = autotrophic beings)


Energy intake
Energy expenditure
Variation in energy reserves
- basal metabolism
+physical activities
+thermogenesis
ensures the continuity of energy expenditure


Energy intake
Energy expenditure
External work
Total metabolic activities
Food
Internal work
•Posture, chivering , respiratory mvt
• Biological activity: (active transport , synthesis reactions, maintenance growth and cellular repair
Energy storage
ENERGY INTAKE AND EXPENDITURE

- In the human body, this chemical energy is usually stored as high-energy molecules ATP, ADP, and CP.

- However, the conversion of chemical energy into work is not perfect: 20% is used for the functioning of the organism, and 80% is lost as heat production

Exemple of functioning

- Mechanical work: muscle contraction
- Electrical work: generation of membrane potentials (pump:

  Na +/K+ ATPase

- Chemical work: synthesis of new molecules

O2

Energy production requires

Organic matter

(food origin)

FOOD

Organic compounds whose degradation produces energy

- Carbohydrates
- Lipids
- Protéines

Non-organic compounds: not degraded for energy

- water , minérals
- Trace elements

II- Basal Métabolism(BM)

- Sum of chemical and biological transformations in the body that maintain and allow organism evolution

  1. Anabolism: synthesis activity of living matter
  2. Catabolism: destruction activity (combustion)

- Basal Metabolism (BM)
- Quantity of energy used to maintain vegetative life, related to the functioning of vital organs (brain, heart, kidney, etc.)
- Measured in Kcal/m^2 of body surface/hour

Basal Métabolism (BM)

- BM excludes energy expenditure related to fighting cold or heat and those not immediately essential for life
- BM is not the lowest metabolic level

sleep

Basal Métabolism (BM)

- Represents 45 to 70% of total energy expenditure = expenditure of organs and tissues (liver 20% - brain 20%)
- BM varies according to: sex, age, body composition

Total energy expenditure decreases with age:

  - BM decreases about 2% every 10 years (due to reduction in lean mass with age)
  - Time and intensity of physical activity decrease with age

Measurement conditions for BM

- Awake subject
- Food restriction for 12 to 16 hours
- Strict rest with muscle relaxation for at least 30 minutes
- Ambient temperature = 21°C for a lightly dressed subject.

MB value and physiological variations

Adult (healthy mal) : BM = 40 Kcal/m²/h

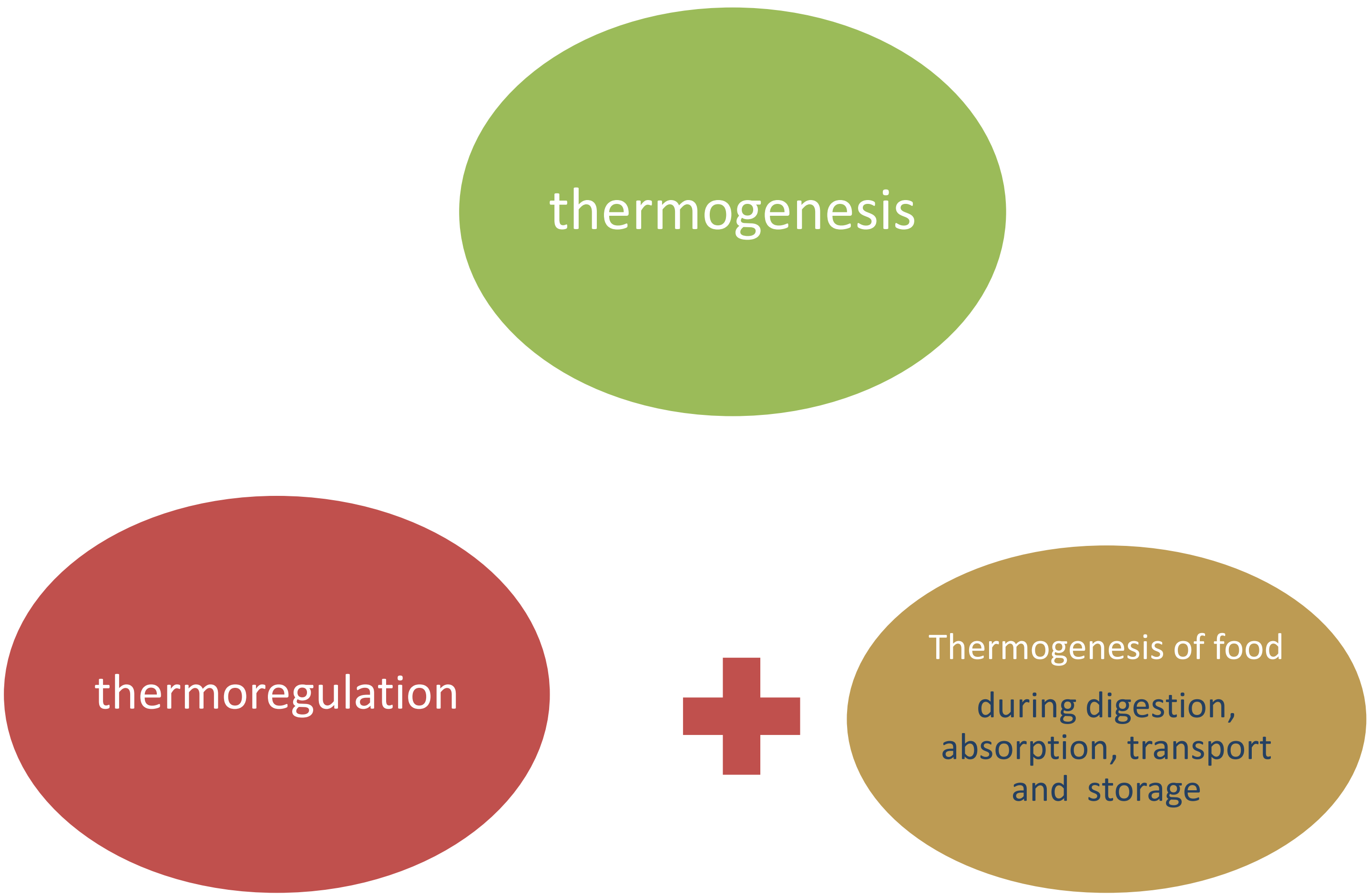
↓ women / men (↑ pregnancy , lactation and ↓ after menopause

○↑ birth ➡ 01 year then ↓ progressively ↑ until puberty.

○ Pathological states that increase BM:

-hyperthyroidism, infection.

III- THERMOGENESIS



THERMOGENESIS

THERMOGENESIS OF FOOD

- -Thermogenesis releases some energy used to maintain body temperature (thermal effect of food or postprandial thermogenesis)

- Represents about 10% of daily energy expenditure

IV- physical activity

Définition

- **<u>sedentariness:</u>**

- State in which movements are minimized and energy expenditure is close to resting metabolism.

__<u>physical activity:</u>__

- Any movement produced by skeletal muscles causing significant energy expenditure above resting levels.

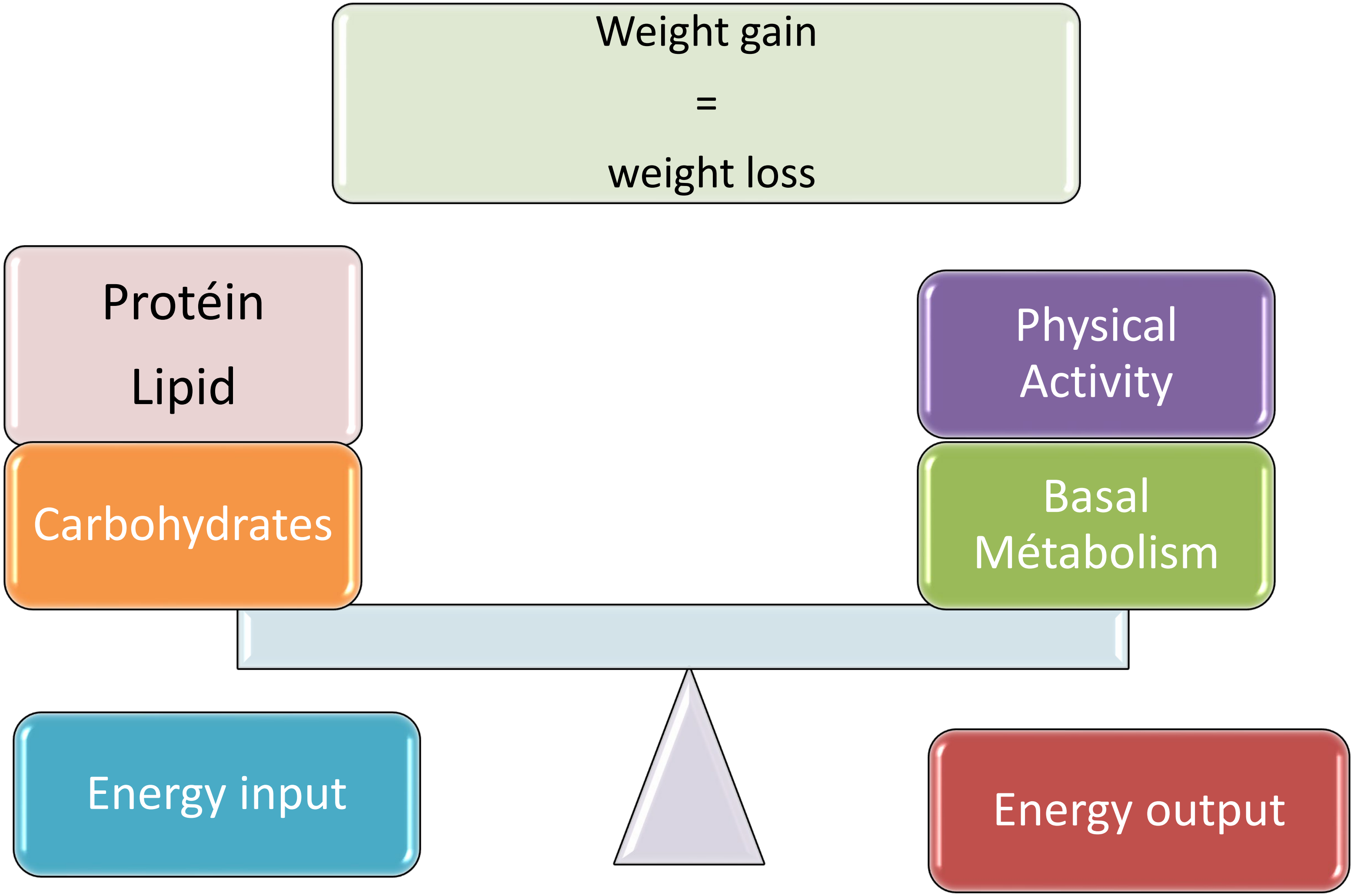
-

IV- physical activity

***sport:**

Physical exercise in the form of individual or team games, often competitive, following specific rules

V- Energy Intake-Expenditure Balance



III-Measurement Methods

Calorimétry:

- Measurement of the amount of energy used by a living organism

- Allows an overall evaluation of its functioning

- Unit of measurement =**Kilocalorie** (Kcal) or **Kilojoule** (KJ)

1Kcal=4.185KJ.

CALORIMETRY

CAN BE

DIRECT

(bomb calorimeter)
or
 isolanted chamber

INDIRECT

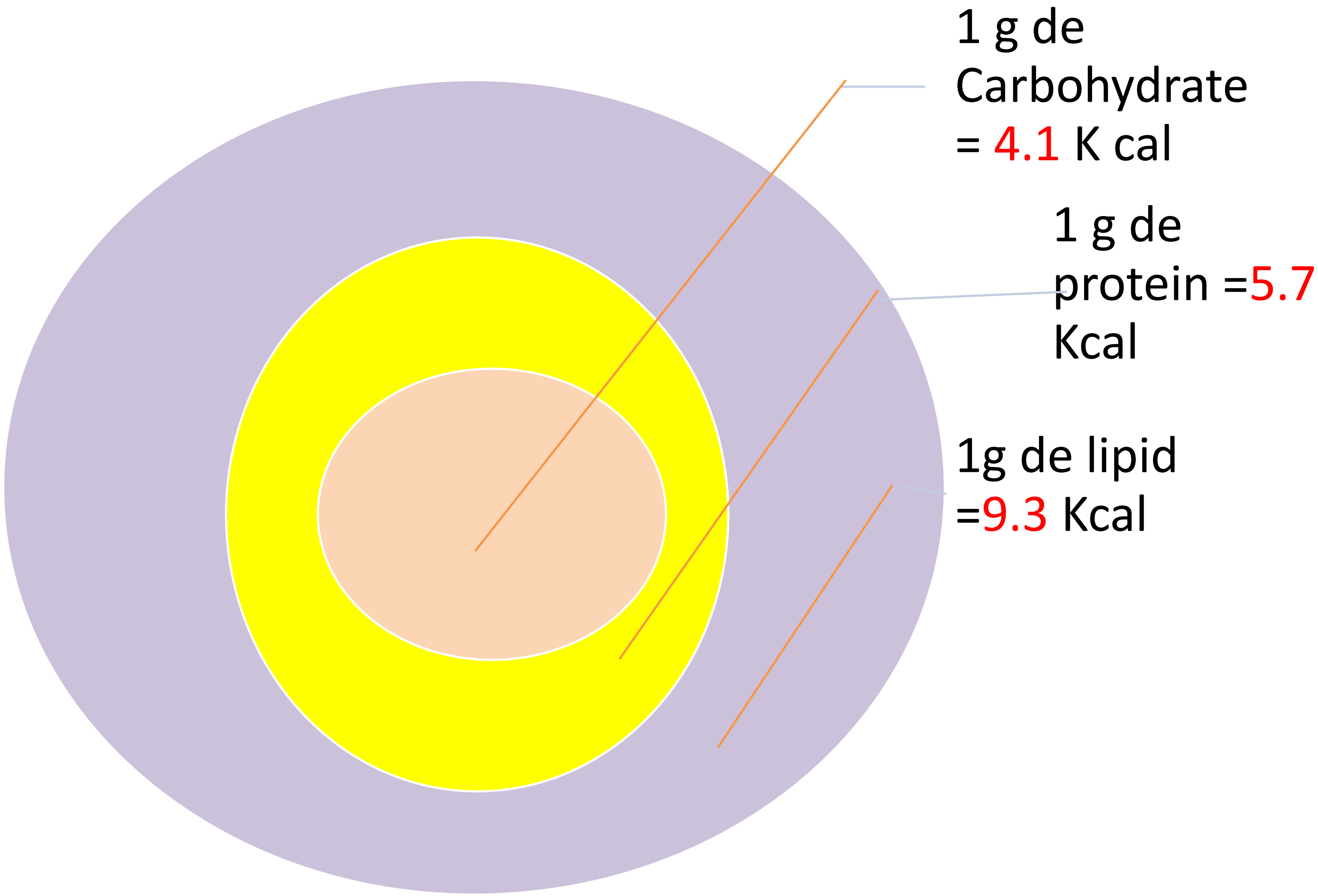
1° dietary or

2° respiratory

CALORIMETERIC BOMB

- Complete combustion of carbohydrates and lipids in presence of O_2 under high pressure produces H_2O, CO_2, heat, and nitrogen for proteins
- Resulting energy = theoretical

Theoretical energy values of nutrients:



(2) **Isolated chamber**

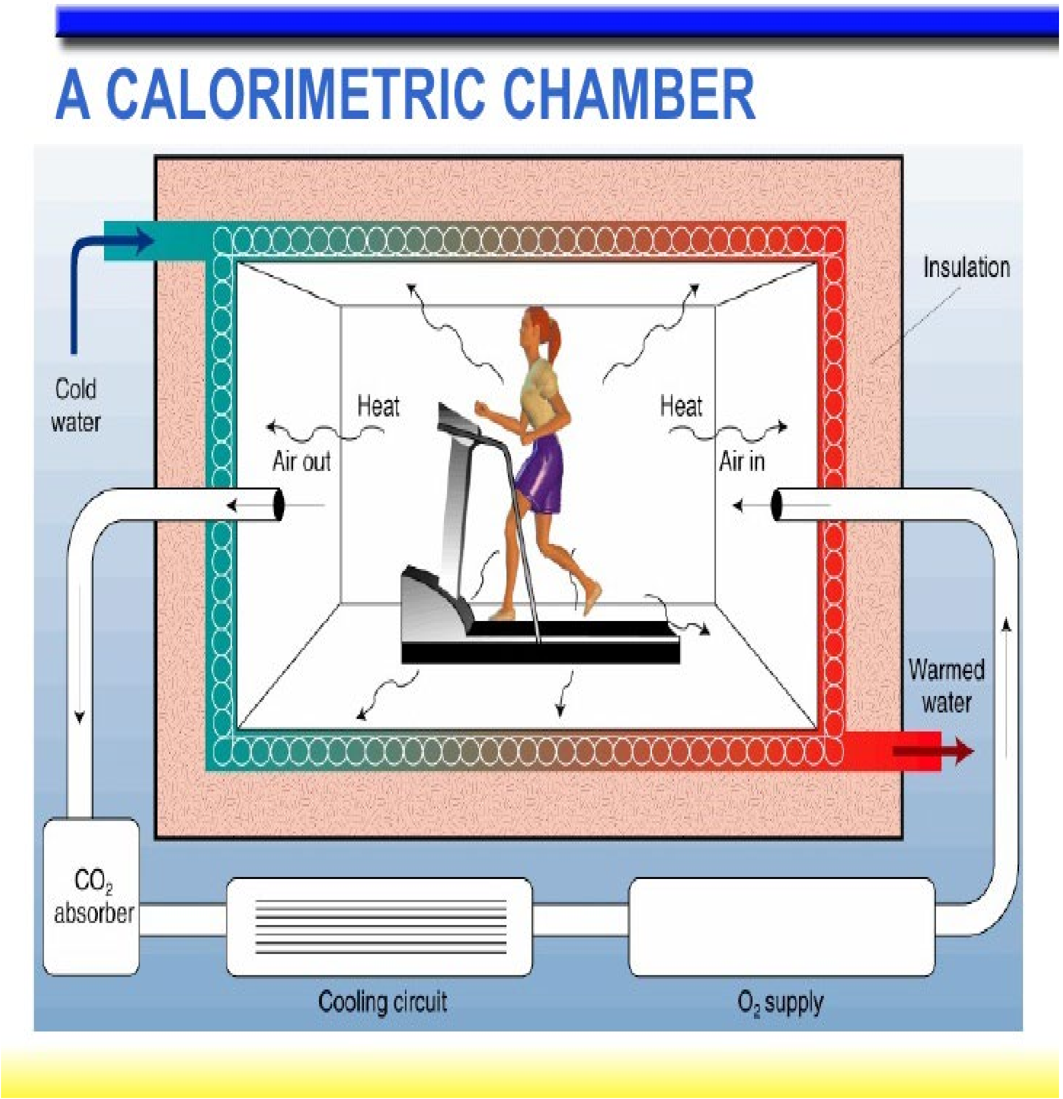
- Objective Measures heat produced by a living organism (thermal metabolism), reflecting its bioenergetics

- Principle: maintain constant temperature of the isolated chamber (adiabatic) using a heat exchanger

Direct Calorimetry

- Subject placed in a chamber with water circulating in the walls
- (**T°s –T°e**) represents the heat produced by the subject and transferred to water

- Expensive and bulky method, rarely used in clinical practice



Indirect calorimétry

1) Dietary method

- Energy value of protein nutrient in the body differs from its theoretical value
- Protein is partially burnt in the body giving a biological or real energy value less than theoretical
- Real energy value of carbohydrates and lipids is approximately equal to theoretical

Real energy values

1 g de carbohydrate =4.1 K cal

1 g de protein = 4.8 Kcal

1 g de lipid = 9.3 Kcal

- Intestinal absorption of nutrients is partial, leading to a practical energy value

Practical energy values

1 g carbohydrate = 4 Kcal

1g protein=

4 Kcal

1 g LIPID = 9 Kcal

- Knowing the weight of each nutrient in the 24-hour food intake (food ration) allows calculation of practical energy intake (caloric ration Q):

Q= 4(G) + 4(P) + 9 (L) Kcal

Indirect calorimétry
2) respiratory method

- Most used in medical practice

- Based on measuring O_2 consumption in a stable state and knowledge of caloric equivalent of nutrient per liter of O_2

FOOD $+O_2 \rightarrow H_2O$+NRJ + CO_2

Caloric equivalent = energy equivalent = thermal coefficient of O_2: energy released per liter of O_2 for nutrient combustion

Thermal coeff of O_2 =NRJ/VO_2

Dont NRJ= 673 Kcal

5.05 Kcal for carbohydrates

4.70 Kcal for lipids

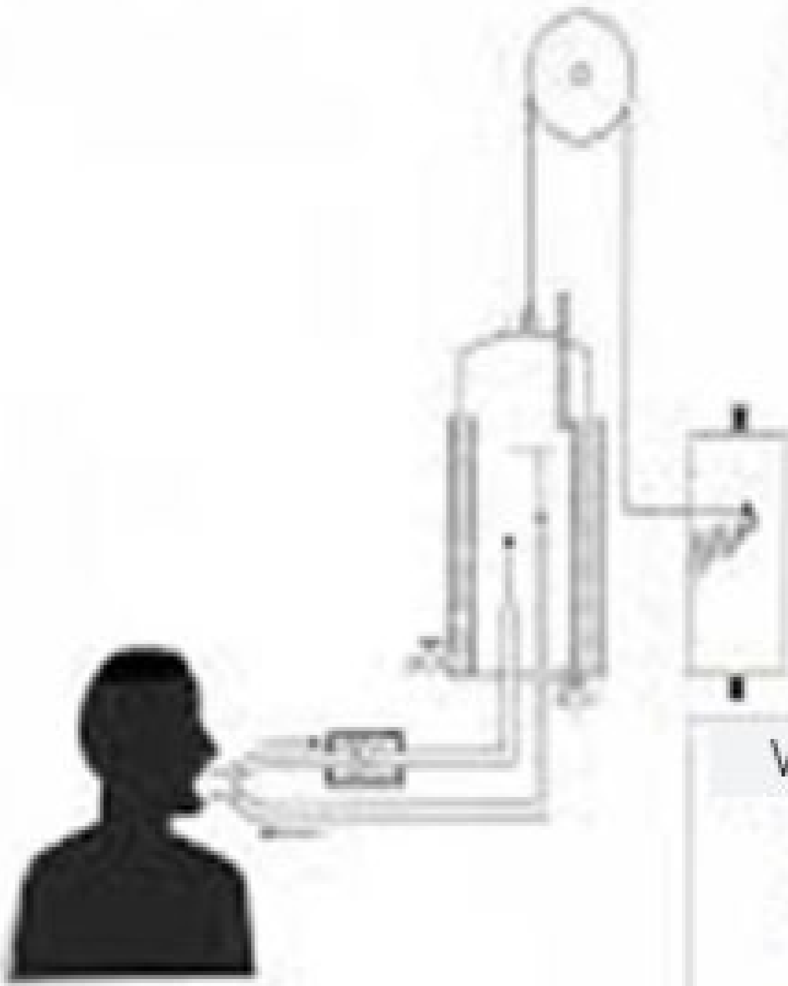
4.70 Kcal for protein

In practice, exact participation of three nutrients during O_2 consumption measurement is difficult to know, so an average caloric equivalent of 4.8 Kcal is used

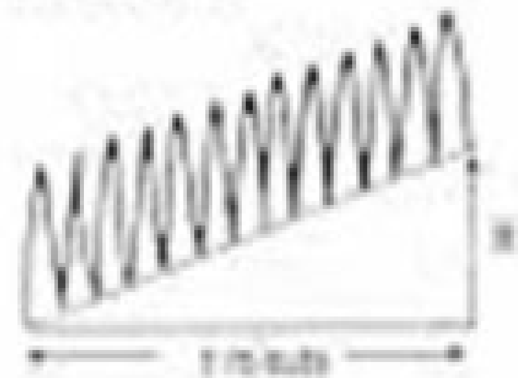
NRJ = 4.85 x VO_2

O2 consumption measurements

a. Closed-loop method (spirometry)



The slope of the route is due to the OG Consumption



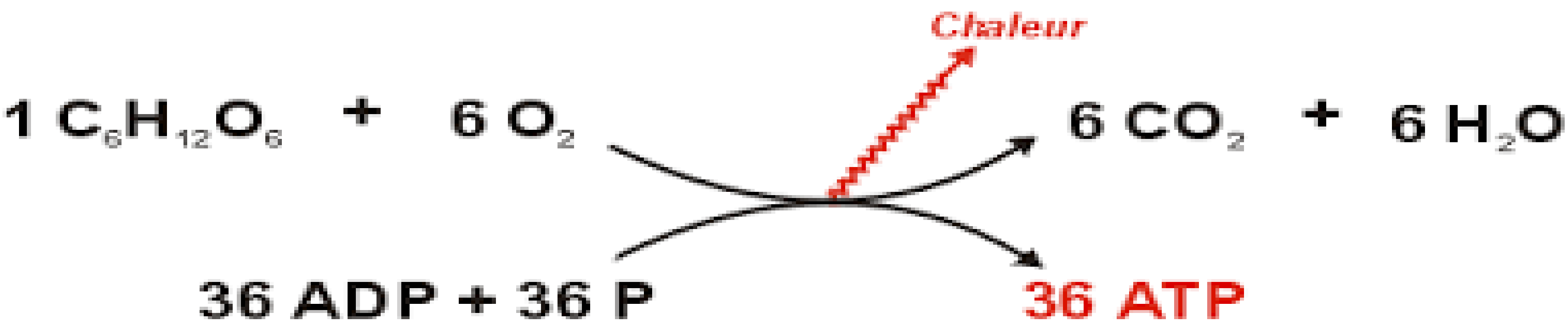
1 cm = 0.2 L

V'O2(L/min) = H (cm min) x 0.2 (1/cm)

D.E- VO2(1/min) x 4.8 (Kcal)

V'O. in STPD conditions

Glucose oxidation



The reaction of the breath can be written as:

602 + CH12 08 ——— * 6 CO2 + 6 H20 + Energy enabling The synthesis of New organic matter

Matière organique : glucose

Respiratory quotient concept

- Ratio R= VCO_2/VO_2

It is calculated after the additional measurement of the amount of CO_2 produced (VCO_2)

Sa valeur est variable selon la nature du nutriment brulé; ainsi elle est égale à :

Lipid =0.7

Protein R = 0.8

carbohydrates =1(idéal)

VII-Food Ration :

Qualitative and quantitative composition of daily diet which must provide:

- Sufficient caloric ration for maintenance (maintenance ration)
- For organism work (work ration)

Maintenance ration: 1600 Kcal for women

Work ration:

500 Kcal for light physical activity

4000 Kcal for very intense activity in extreme cold conditions

Food ration : définition 2

Daily quantity and quality (nature) of food ingested by a subject

Provides energy, water, minerals, and vitamins in sufficient quantity to meet body needs

Food : any substance that can serve as nutrition for a living being, containing nutrients or ingredients beneficial physiologically.

Nutrient : any organic or inorganic compound in food usable by the body

FOOD RATION COMPOSITION

Approximately **30** to **50 g** = **15%** protein for men, more for women during pregnancy or lactation

- Proteins are mainly needed for structure and to cover nitrogen losses estimated at 2.5 g/day

Nitrogen losses pathways and values

	urinary	**fécal**	**cutaneous**	**secretions**
NITROGEN LOSSES (g/24 h)	1.4	0.4	0.13	0.08

Total = 2 g per day

Composition of food ration

Lipid needs = 30-35% of caloric ration

Carbohydrate needs = 50-55% of caloric ration

(Minimum carbohydrate need = 150 g/day)

In total

Typical ration: 55% carbohydrate, 30% lipid, 15% protein
For a 2400 Kcal/day ration

Different food groups::

According to their properties and nutritional qualities, foods are grouped into six or seven categories. A balanced diet will draw from each of the following food groups :

Group	FOOD	compositions
1	Meat, fish, eggs, poultry, offal	animal proteins, minerals, vitamins A, B, D
2	Dairy products (milk, yogurt, cheese	animal proteins, lipids, minerals, vitamin A
3	Fruits and vegetables, oleaginous fruits	carbohydrates, water, minerals, vitamins C, E, vegetable proteins, lipids
4	Cereals, bread, rice, starches, potatoes	vegetable proteins, carbohydrates
5	Fats: visible (butter, oil), invisible (milk, peanuts):	Lipids , vitamin A
6	Sugar and sweet products (chocolate, honey, jam	carbohydrates
7	Water, unsweetened coffee and tea, fruit juices, sodas, Coca-Cola	0 Kcal carbohydrates